The process of transforming the C-level models description to technology-dependent netlist is performed by the high-level synthesis tool.

Placement

The physical implementation of the ASIC is initiated by the placement tool. With the aid of a 2-D floorplan that's provided by the ASIC designer, locations for each gate in the netlist are assigned by the placer tool. The result would be a place gates netlist.

This netlist would include the physical location of all the standard cells in the netlist. However, it does have an abstract description of the way the gate's terminals are wired to each other.

Usually, the standard cells would feature a constant size in at least one dimension. This would enable them to be lined up in rows on the integrated circuit. The chip would contain a large number of rows. The power and ground would run next to each row. Each of the rows would be filled with the different cells which make up the actual design.

Placers follow some rules as well. For instance, a unique location is assigned to each gate on the die map. Once a given gate is placed, it shouldn't occupy or overlap the location of another gate.

Routing

The router adds the signal connect lines and the power supply lines using both the placed-gates netlist and the library's layout view. The completely routed physical netlist would have the listing of the gates from synthesis, each gate's position from placement and from rioting, the drawn interconnects.

The verification processes

Two verification processes that are used are Design Rule Check(DRC) and Layout Versus Schematic(LVS). A rigid observance of transistor spacing, power density rules and metal layer thickness is needed for device fabrication at deep-submicrometer(0.13 0.13 µm and less). With DRC, an exhaustive comparison of the physical netlist against a set of foundry design rules happen. If any violations are noted, they are flagged.

With the LVS process, it can be confirmed that the layout indeed has the same structure as the associated schematic. Usually, this forms the last step in the layout process.

A schematic diagram as well as the view extracted from a layout form the input of the LVS tool. Then, a netlist is generated from each one, after